

Team Project Midterm Report: Spam Email Detection

1. Data

Dataset: UCI Spambase Dataset Number of Instances: 4,601 emails Attributes: 57 continuous attributes (word frequencies, punctuation counts, capital letter statistics), plus 1 class label (spam = 1, not spam = 0). Type of Data: Text features extracted numerically. Preprocessing: • Removed duplicate and missing entries. • Normalized attribute values (scaling between 0–1). • Split dataset into training (80%) and testing (20%). Attributes Used: All features except the email ID column (not relevant for prediction).

2. Data Mining Task

Task: Classification — predict whether an email is spam or not spam. Goal: Compare the performance and efficiency of multiple machine learning algorithms for this binary classification problem.

3. Progress

Algorithms Tested: • Naive Bayes (baseline) • Logistic Regression • Decision Tree Classifier • Random Forest Classifier Preliminary Results: • Naive Bayes: ~89% accuracy • Logistic Regression: ~92% accuracy • Decision Tree: ~90% accuracy • Random Forest: ~94% accuracy (best performance so far) Challenges: • Feature correlation between word frequencies causes slight overfitting for tree-based models. • Balancing between false positives and false negatives — spam detection requires minimizing false negatives for safety.

4. Schedule

Next Steps: • Apply cross-validation for more reliable evaluation. • Try Support Vector Machine (SVM) for comparison. • Fine-tune hyperparameters (e.g., number of trees in Random Forest). • Create performance visualization charts (ROC curve, confusion matrix). • Begin drafting final report and preparing presentation slides.